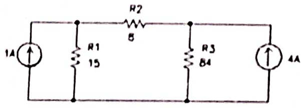
Node and Mesh analysis Numerical

Problems for Mesh & Nodal Analysis
 # 1.

Q1

Figure 1 - Problem 1
Find the voltage across R2.



Answer : Answer: VR2 = 24v with right side more positive

Nodal Equations

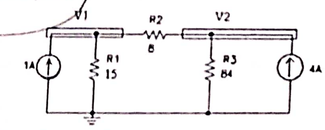


Figure 1 - Problem 1

Node 1	-1A + V1/15 + (V1 - V2)/8 = 0
Node 2	(V2 - V1)/8 + V2/64 -4A = 0
	Answer: VR2 = 24v (right more +)

Mesh Equations

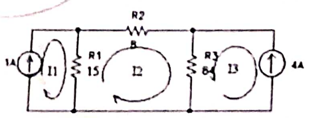


Figure 1 - Problem 1

Loop 1	I1 = 1A
Loop 2	15(I2 - I1) + 8*I2 + 84*(I2 + I3) =0
Loop 3	I3 = -4A
	Answer: VR2 = 24v (right more +)

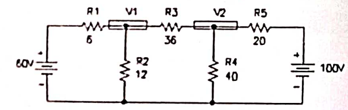


Figure 2 - Problem 2

 # 2. Find the node voltages V1 and V2.

Nodal Equations

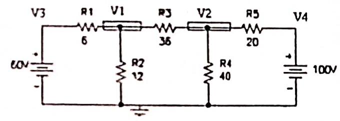


Figure 2 - Problem 2

Node 1	(V1 - V3)/6 + V1/12 + (V1 - V2)/36 = 0
Node 2	(V2 - V1)/36 + V2/40 + (V2 - V4)/20 =0
Node 3	V3 = 60
Node 4	V4 = 100

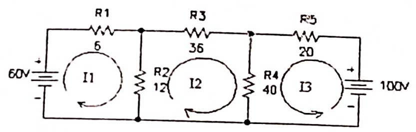


Figure 2 - Problem 2

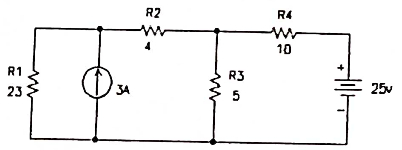
Loop 1 -60v + 6*I1 + 12*(I1 - I2) = 0

Loop 2 12*(I2 - I1) + 36*I2 + 40*(I2 + I3) = 0

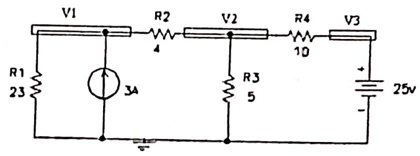
Loop 3 -100v + 20*I3 + 40*(I3 + I2) = 0

Answer: V1 = 42v, V2 = 60v

3. Find the current through R2 and R3



Nodal equations:

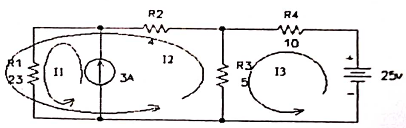


Node 2: (V2 - V1)/4 + V2/5 + (V2 - V3)/10 = 0

Node 3:V3 = 25v

Answer: IR2 = (V1 - V2)/4 = 2A, IR3 = V2/5 = 3A

Mesh equation



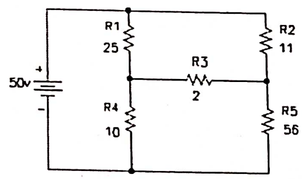
Loop 1 I1 = 3A

Loop 2 5*(I2 - I3) + 4*I2 + 23*(I2 + I1) = 0

Loop 3 -25v + 10*I3 + 5*(I3 - I2) = 0

Answer: IR2 = (V1 - V2)/4 = 2A, IR3 = V2/5 = 3A

4. Find the voltage across R3



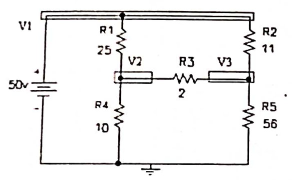


Figure 4 - Problem 4

Node 1 V1 = 50v

Node 2 (V2 - V1)/25 + V2/10 + (V2 - V3)/2 = 0

Node 3 (V3 - V1)/11 + V3/56 + (V3 - V2)/2 = 0

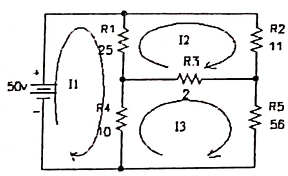


Figure 4 - Problem 4

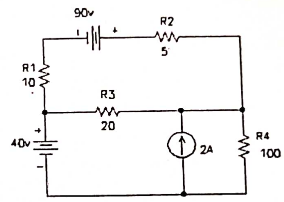
Loop 1 -50v + 25*(I1 - I2) + 10*(I1 - I3) = 0

Loop 2 25*(I2 - I1) + 11*I2 + 2*(I2 -I3) = 0

Loop 3 10*(I3 - I1) + 2*(I3 - I2) + 56*I3 = 0

Answer: VR3 = V3 - V2 = 3v

 # 5.Find the current through R3.



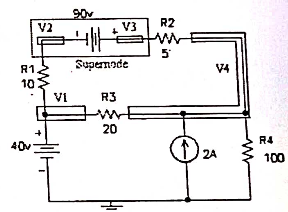


Figure 5 - Problem 5

Node 1 V1 = 40v

Supernode 1 V2 + 90 = V3

Supernode 2 (V2 - V1)/10 + (V3 - V4)/5 = 0

Node 4 (V4 - V3)/5 + V4/100 - 2A + (V4 - V1)/20 = 0

Answer: IR3 = (V4 - V1)/R3 = 3A

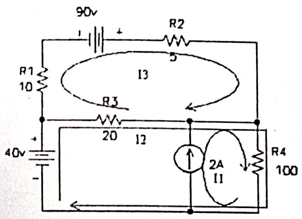


Figure 5 - Problem 5

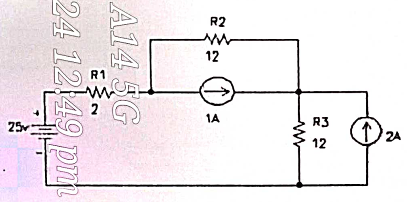
Loop 1 I1 = 2A

Loop 2 -40v + 20*(I2 - I3) + 100*(I2 + I1) = 0

Loop 3 10*I3 - 90v + 5*I3 + 20*(I3 - I2) = 0

Answer: IR3 = (V4 - V1)/R3 = 3A

6. Find the voltage across the 1A current source



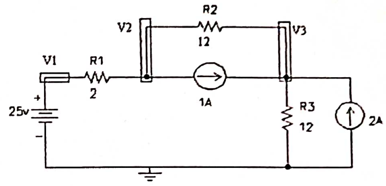


Figure 6 - Problem 6

Node 1 V1 = 25v

Node 2 (V2 - V1)/2 + (V2 - V3)/12 +1A = 0

Node 3 (V3 - V2)/12 - 1A + V3/12 - 2A = 0

Answer: V(1A source) = V3 - V2 = 6v

Mesh analysis

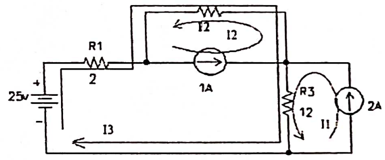


Figure 6 - Problem 6

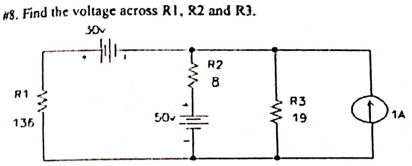
Loop 1 I1 = 2A

Loop 2 I2 = 1A

Loop 3 -25v + 2*I3 + 12*(I3 - I2) + 12*(I3 + I1) = 0

Answer: V(1A source) = VR2 = 12*(I2 - I3) = 6v

#8. Find the voltage across R1, R2 and R3.



Nodal analysis:

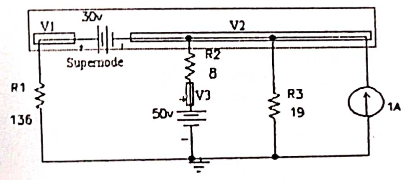
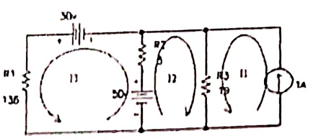


Figure 8 - Problem 8

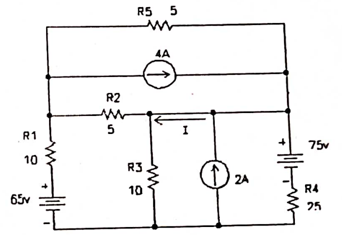
Answer: $V_{R1} = 68v$; $V_{R2} = 12v$; $V_{R3} = 38v$

Node 1	$V1 - 30 = V2$
Supernode	$V1/136 + (V2 - V3)/8 + V2/19 - 1A = 0$
Node 3	$V3 = 50v$

Mesh analysis:



#9. Find the current labeled "I"



Nodal analysis:

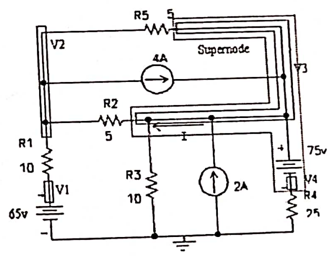


Figure 9 - Problem 9

Node 1	$V1 = 65v$
Node 2	$(V2 - V1)/10 + (V2 - V3)/5 + (V2 - V3)/5 + 4A = 0$
Node 4	$V4 + 75 = V3$

Mesh analysis:

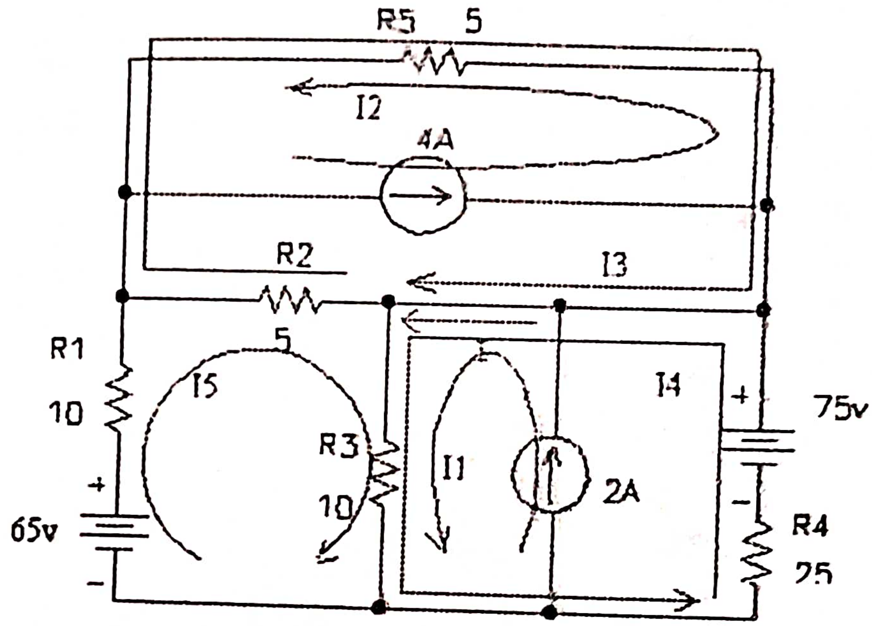


Figure 9 - Problem 9

Loop 1	$I1 = 2A$
Loop 2	$I2 = 4A$
Loop 3	$5*(I3 - I5) + 5*(I3 - I2) = 0$
Loop 4	$25*I4 - 75v + 10*(I4 + I1 + I5) = 0$
Loop 5	$-65v + 10*I5 + 5*(I5 - I3) + 10*(I5 + I1 + I4) = 0$

Answer: $I = I1 + I4 - I3 = 6A$

Galaxy A14 5G
30 April 2024 12:50 pm

Numerical based on kvl or kcl

Kirchhoff's Voltage law: The algebraic sum of all the voltages in any mesh (closed circuit) is zero.

Determination of signs while solving problems-

1. **Voltage Source:** A positive sign is assigned to the EMF when going from the negative terminal of the voltage source to the positive terminal since there is an increase in potential. On the other hand, when moving from the positive terminal to the negative terminal, potential drops and the EMF is marked as negative.

(a) Rise in Voltage (b) Fall in voltage

2. **Resistance:** The potential drop is negative when passing through the resistance in the same direction as the current flow because the potential decreases. However, if the resistance is traversed in the opposite direction to the current flow, the potential increases, resulting in a positive voltage drop.

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(a) Fall in voltage (b) Rise in voltage

Note:

1. Consider the second polarity of the voltage source / battery and resistance in the direction of current while solving the problems.
2. To simplify calculations, it is advisable to always assume that the current is flowing clockwise.

Q1 **Find the power consumed by the combination of resistances when connected to a voltage source of 100 V at terminals *P-Q* (Fig. 1.9).**

(*Ans:* 3.125 kW)

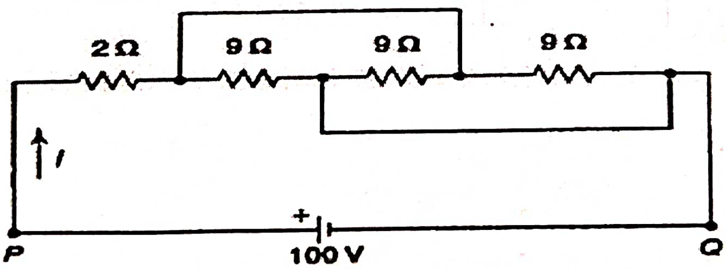
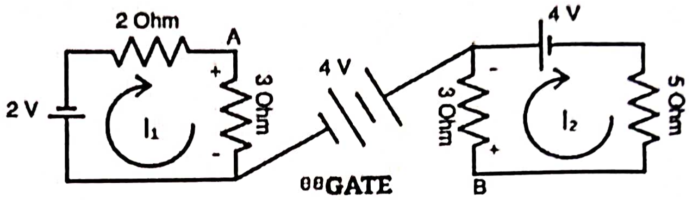


Fig. 1.9

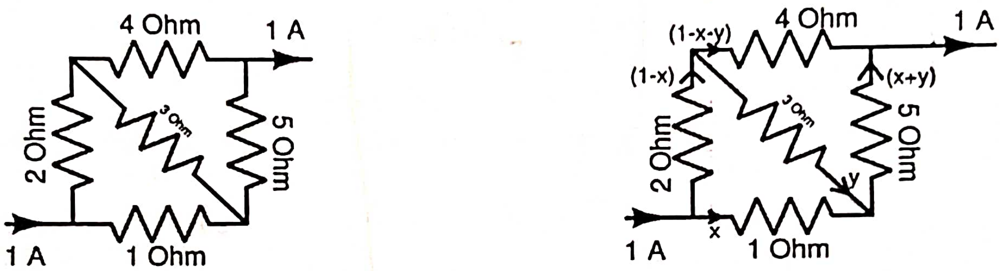
[*Hint:* Fig. 1.9 can be reduced as shown below:

Numerical based on kvl or kcl

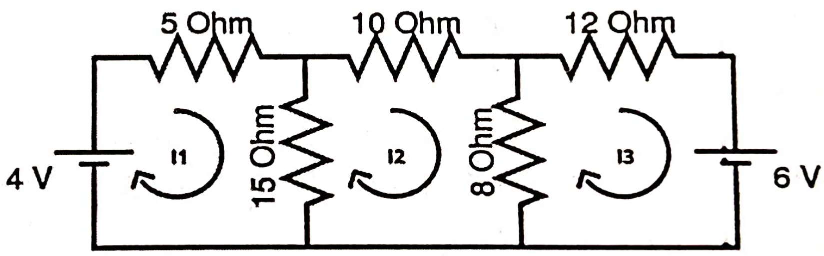
Q2. **Find potential difference between X and Y.**



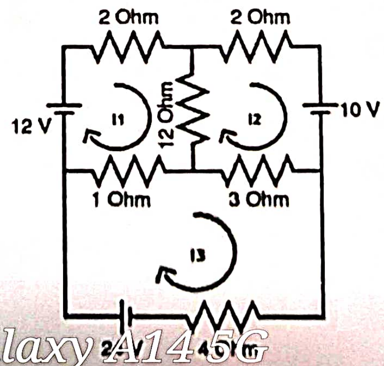
Q3 **Find current flowing through all resistors.**



Q4 **By using KVL find Current flowing through 10 Ω resistance.**

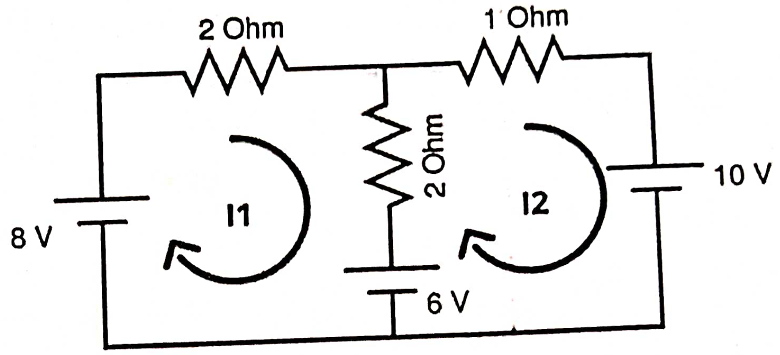


Q5 **By using Kirchhoff's voltage law (KVL) / Mesh analysis find the current flowing through a 4 Ω resistor.**

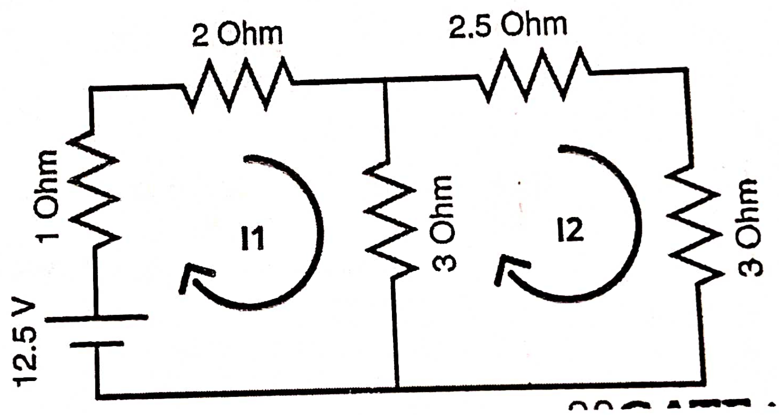


Numerical based on kvl or kcl

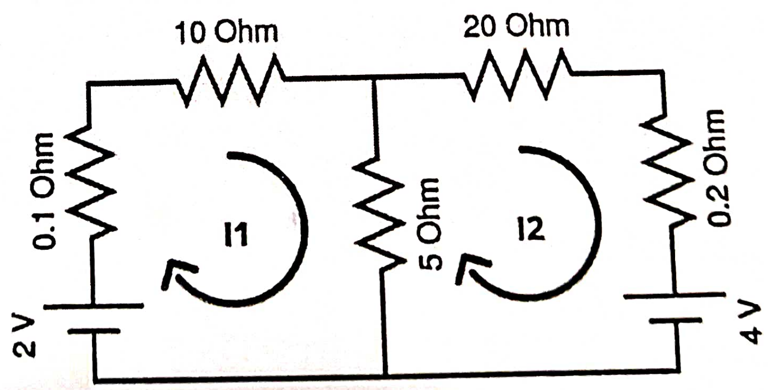
Q6 By using Kirchhoff's voltage law (KVL) / Mesh analysis find the current flowing through a 1 Ω resistor.



Q7 By using Kirchhoff's voltage law (KVL) / Mesh analysis find the current flowing through all resistances.

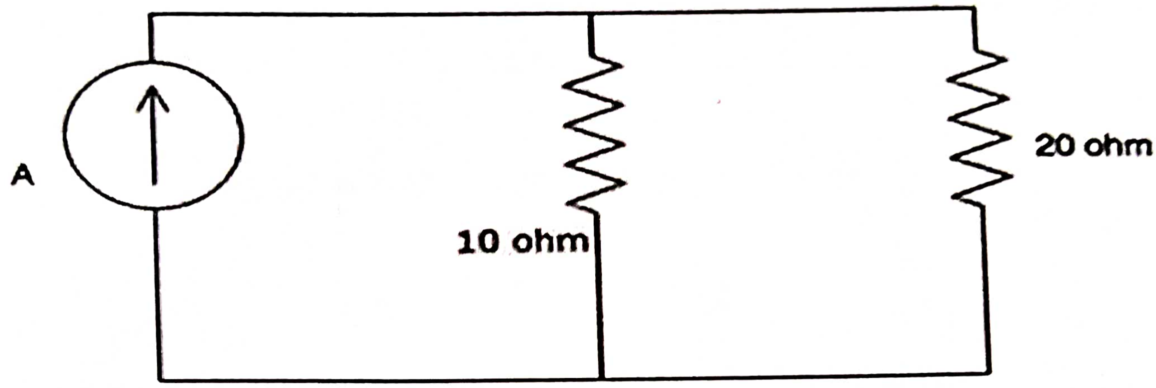


Q8 By using Kirchhoff's voltage law (KVL) / Mesh analysis find the current flowing through a 5 Ω resistor.

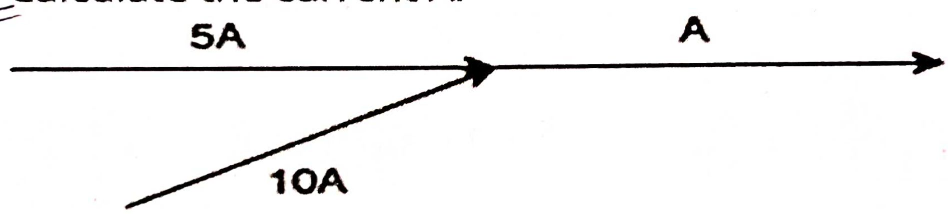


Numerical based on kvl or kcl

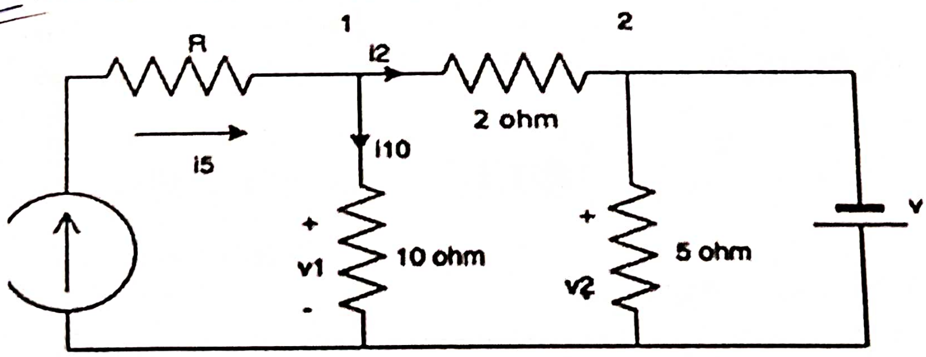
Q9 Calculate the current across the 20 ohm resistor.



Q10 Calculate the current A.



Q11 Find the value of v if v1=20V and value of current source is 6A.



Numerical based on kvl or kcl

Q12 Find the equivalent resistance between points a and b of the infinite ladder shown in Fig. 1.13. (Ans: $R_{eq(a-b)} = \frac{1+\sqrt{5}}{2} \cdot R$)

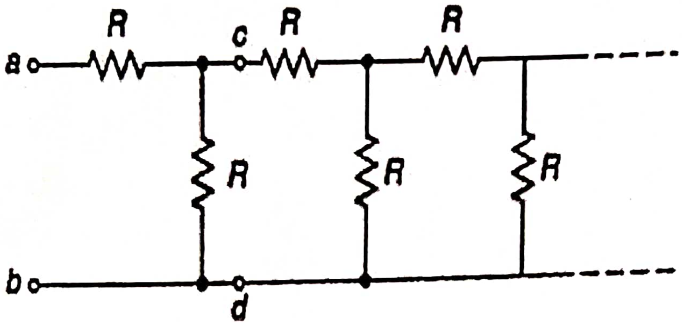


Fig. 1.13

Q13 KCL is applied at __________

Loop

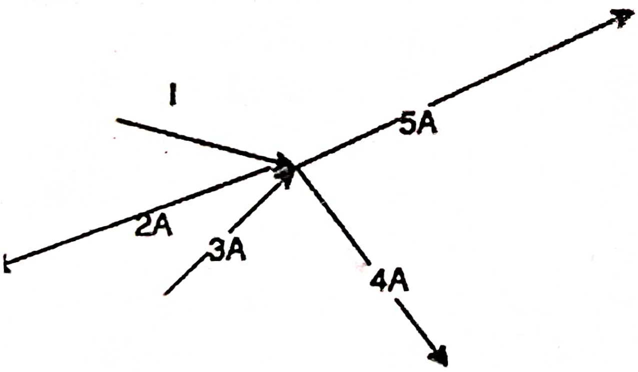
Node

Both loop and node

Neither loop nor node

View Answer

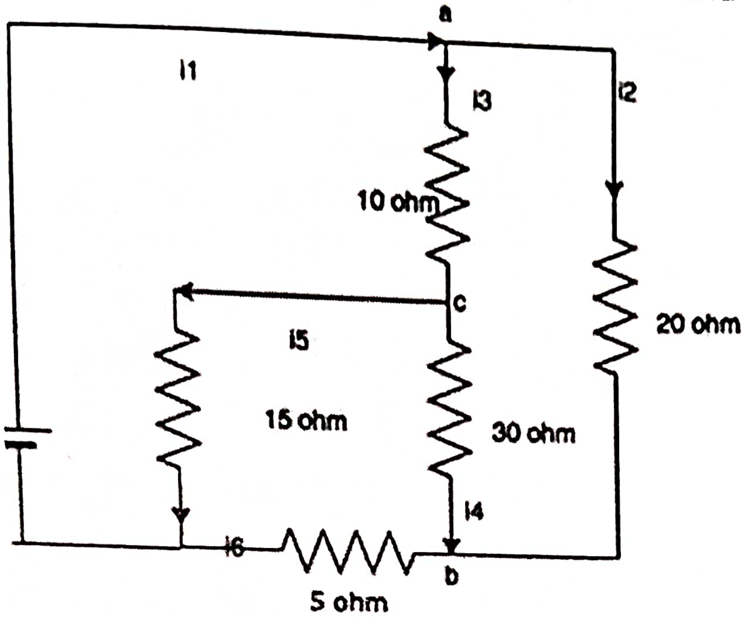
Q14 What is the value of the current I?



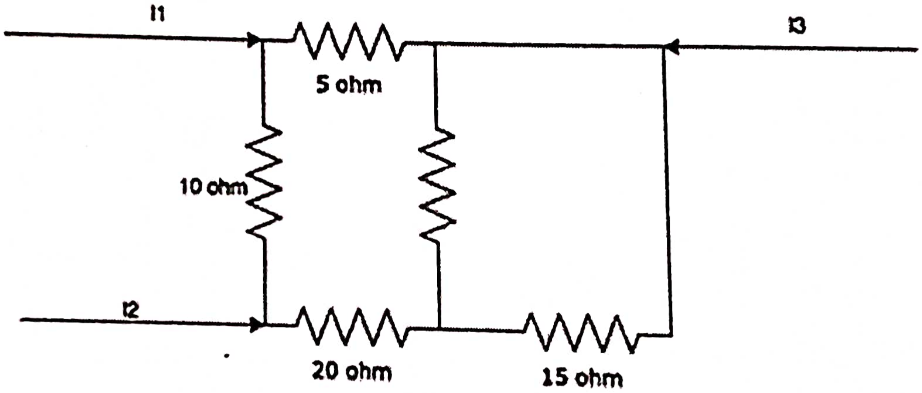
$I+3 = 5A+4A$

Numerical based on kvl or kcl

Q15 Find the value of i2, i4 and i5 if i1=3A, i3=1A and i6=1A.

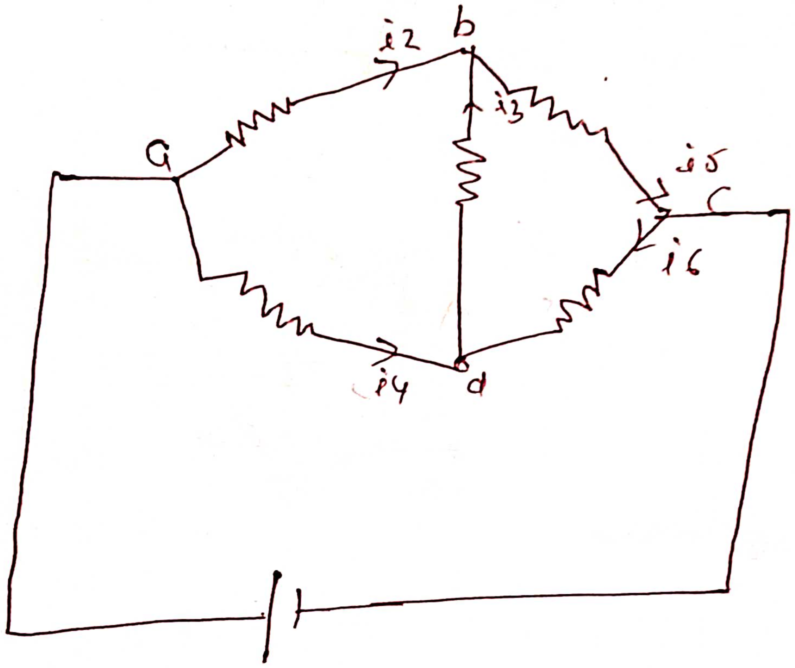


Q16 Calculate the value of I3, if I1= 2A and I2=3A.



Q19 find the current unknown i_4, i_3, i_6 given

$i_1 = 20A$ $i_2 = 12A$ $i_5 = 8A$



Q